RESEARCH ARTICLE

Neurovascular Unit Dysfunction and Cerebral Small Vessel Disease in Parkinson Disease

A Systematic Review and a Meta Analysis

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Abstract

Background and Objectives

Parkinson disease (PD) pathogenesis remains incompletely understood; beyond nigrostriatal loss, nondopaminergic mechanisms including neurovascular unit dysfunction may contribute to disability. Cerebral small vessel disease (CSVD) burden reflects neurovascular dysfunction in the form of white matter hyperintensities (WMH), lacunes, cerebral microbleeds (CMB), and enlarged perivascular spaces (ePVS). In PD cohorts, CSVD burden correlates with worse motor and gait scores. The aim of the study was to explore whether patients with PD exhibit greater CSVD burden than healthy controls (HC).

Methods

We conducted a PRISMA-conformant systematic review and meta-analysis of studies including adults with idiopathic PD and a HC group that presented data comparing CSVD burden in these 2 groups. Six databases (MEDLINE, Embase, CINAHL Plus, CENTRAL, Scopus, and Web of Science) were searched on May 14 and 15, 2024. Two reviewers independently screened records, extracted data, and assessed risk of bias, with discrepancies resolved by consensus. Continuous outcomes were pooled as standardized mean differences (SMD); dichotomous outcomes as odds ratios (ORs). We evaluated small-study effects for pooled analyses with 10 or more studies using funnel plots and Egger regression test.

Results

We examined 13,403 records. Forty-six studies (45 cross-sectional) met inclusion criteria, totaling 3,817 PD and 2,593 HC (mean ages: 66.9 and 66.5, respectively). WMH volume (k = 21) was higher in PD (SMD 0.36, 95% CI 0.11–0.62). Visual ratings also indicated higher WMH in PD: global (k = 14) SMD 0.27 (95% CI 0.08–0.46); periventricular (k = 11) SMD 0.32 (95% CI 0.12–0.51); deep (k = 8) SMD 0.20 (95% CI 0.09–0.31). Differences in CMB (k = 6; OR 1.18, 95% CI 0.57–2.42) and lacunes (k = 4; OR 1.48, 95% CI 0.58–3.78) were not significant. ePVS results were heterogeneous but trended toward greater burden in PD, most notably in the midbrain (k = 3; SMD 1.80, 0.15–3.44). Overall evidence quality was rated as low, reflecting the observational nature of the included studies.

Discussion

Our analysis showed PD to be associated with greater WMH burden and increased midbrain ePVS. Pooled differences in CMB and lacunes were not significant. Substantial heterogeneity and cross-sectional designs limit certainty; standardized imaging and prospective cohorts are needed to define mechanisms and clinical implications.

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Glossary

BBB = blood-brain barrier; **CMB** = cerebral microbleeds; **CSVD** = cerebral small vessel disease; **dWMH** = deep WMH; **ePVS** = enlarged perivascular spaces; **HC** = healthy controls; **NVC** = neurovascular coupling; **NVU** = neurovascular unit; **OR** = odds ratio; **PD** = Parkinson disease; **PRISMA** = preferred reporting items for systematic reviews and meta-analyses; **pvWMH** = periventricular WMH; **RoB** = risk of bias; **SMD** = standardized mean differences; **VRS** = visual rating scales; **WMH** = white matter hyperintensities.

Introduction

Neurodegenerative and neurovascular diseases are increasingly recognized as related entities.[1,2] Although historically considered distinct, the neurovascular unit (NVU) model has gained traction over the last 2 decades after its introduction at the 2001 Stroke Progress Review Group meeting of the National Institute of Neurological Disorders and Stroke.[1] This model emphasizes the interdependence of cerebral vasculature, neurons, and glia in maintaining brain homeostasis and brings to the forefront the role the brain's vasculature plays in neuronal and glial activity. It also underscores how vascular dysfunction may contribute to neurodegenerative disease progression, even in the absence of overt ischemic injury.[2]

Environmental conditions in the brain are tightly regulated to maintain homeostasis. The NVU supports this through neurovascular coupling (NVC), the blood-brain barrier (BBB), and glymphatic clearance. NVC coordinates local cerebral blood flow with regional neuronal activity, and its impairment may lead to chronic microvascular injury. The BBB forms a selective interface that regulates molecular exchange and protects neural tissue from circulating toxins and inflammatory mediators. The glymphatic system facilitates bulk clearance of interstitial solutes, including protein byproducts, and its dysfunction is thought to be reflected radiographically by enlarged perivascular spaces.[3] Together, these interconnected components of the NVU maintain vascular, metabolic, and immunologic stability within the brain.

This model has been used to clarify biological processes relevant to Parkinson disease (PD) progression.[2] PD is the fastest growing neurologic disorder, having doubled to over 6 million from 1990 to 2015, and is expected to double again by 2040.[4] There are no currently available medications that slow the progression of disease, with exercise as the only known disease modifying intervention for PD, although its mechanisms are unclear.[5] Research into PD pathogenesis provides hopes of delineating potential disease modifying therapies and defining which outcome measures will be most meaningful. In this context, understanding the role of the neurovascular unit in PD is increasingly crucial.

NVU disruption in Parkinson disease often manifests as cerebral small vessel disease (CSVD) on MRI, characterized by enlarged perivascular spaces (ePVS), white matter hyperintensities (WMH), cerebral microbleeds (CMB), and lacunes. Clinically, CSVD is linked to increased risk for dementia, depression, urinary incontinence, and gait dysfunction.[6] In line with these known sequelae of CSVD, multiple studies report that patients with PD with CSVD exhibit worse motor impairment[7-10] and greater cognitive impairment.[8,11-13] An unanswered question remains to what extent NVU disruption contributes to CSVD in PD. This review critically evaluates current evidence linking PD and imaging features of CSVD.

Methods

Eligibility Criteria

Study selection occurred in 2 phases: title/abstract screening and full-text review. During title/abstract screening, we excluded studies in which subjects explicitly did not have idiopathic PD (e.g., atypical or secondary parkinsonism), those with no contemporaneous healthy control group, those with inadequate imaging to assess various markers of CSVD (e.g., no FLAIR/T2 for WMH; no T2*/SWI for CMB), or those that were case reports/series, narrative reviews, or editorials. The remaining studies proceeded to full-text eligibility review. Studies passed full-text review if the full manuscript was available, the study population consisted of adults with stated PD and a contemporaneous healthy control (HC) group, and at least one CSVD marker was reported with extractable PD-HC comparison: WMH, lacunes, cerebral microbleeds, enlarged perivascular spaces, or total CSVD burden (e.g., means/SDs, medians/IQRs convertible, or proportions).

Information Sources and Search Strategy

We developed the methods for this systematic review by following the Methodological Expectations of Cochrane Intervention Reviews Manual and then reported the results to align with the preferred reporting items for systematic reviews and meta-analyses (PRISMA). The review protocol was developed a priori but was not registered in PROSPERO or any similar registry.

The clinical research librarian (L.H.) from Louis Calder Memorial Library conducted the search strategy upon consulting with the study team. The Medline (OVID) search strategy was reviewed by another medical librarian (DG) using the Peer Review for Electronic Search Strategies tool.

The initial search strategy was implemented in the Ovid Medline database using both the Medical Subject Headings (MeSH) and keyword variants for "cerebral small vessel disease" and "Parkinson disease." Searches were then adapted for the following databases in addition to Ovid MEDLINE(R) ALL: Embase (Elsevier[14]), Cumulative Index to Nursing and

Allied Health Literature (CINAHL), Plus (Ebsco), Cochrane Central Register of Controlled Trials (CENTRAL) (Cochrane Library, Wiley), Scopus (Elsevier), and the Web of Science platform (Clarivate: Science Citation Index Expanded, Social Sciences Citation Index, Arts & Humanities Citation Index, Conference Proceedings Citation Index-Science, Conference Proceedings Citation Index-Social Science & Humanities, and Emerging Sources Citation Index).

The Medline search strategy was revised for the rest of the databases according to the syntax of each, taking into consideration the utilization of controlled vocabulary such as the following: EMTREE terms for Embase, CINAHL Subject Headings for CINAHL Plus, and Medical Subject Headings for Cochrane Library in the controlled vocabulary parts of the search strategy and the search field, the truncation, and proximity in the keyword parts of the search strategy. All databases were searched on May 14 and 15, 2024, with no restriction on language, date, or other limitations. For the full search strategies, see eAppendix 1. A targeted PubMed update was performed after completion of analysis on November 26, 2025 and identified 727 new records; one study met inclusion criteria, and its findings were incorporated qualitatively in the Results.

Search results were imported into Covidence Web-based software to remove duplicates, title-abstract screening, full-text screening, and data extraction. For Medline (OVID), the results were imported to EndNote 20, exported into RIS output style format, and then imported to Covidence.

Study Selection

Search results were managed in Covidence and deduplicated. Two independent reviewers (drawn from a pool of 8) screened titles/abstracts and then full texts against prespecified eligibility criteria; disagreements were resolved by consensus, with third-reviewer adjudication as needed. Reasons for full-text exclusion were recorded, and when multiple reports from the same cohort were found, the earliest published numbers were collated to avoid double counting. The selection process is summarized in a PRISMA 2020 flow diagram (Figure 1).

Data Collection

Two reviewers independently and in duplicate extracted data using a standardized, prepiloted forms in Excel; disagreements were resolved by consensus. Items extracted included (1) study characteristics, (2) sample characteristics, (3) imaging & rating characteristics, (4) CSVD outcomes, and analysis details. Details on item extraction are provided in eAppendix 2.

Risk of Bias Assessment

Two reviewers independently assessed study quality using a modified Newcastle-Ottawa Scale for cross-sectional studies, evaluating sample selection, comparability, and outcome measurement.[15] Studies scoring <6 of 9 were considered at least moderate risk of bias (eFigures 1–6).

Synthesis Methods

Meta-analyses were performed on cross-sectional studies comparing cerebral small vessel disease (CSVD) markers in patients with idiopathic PD vs HC.

Statistical Model and Heterogeneity

We performed meta-analyses using both a fixed-effect model and a random-effects model with the DerSimonian-Laird estimator for between-study variance (τ^2). For continuous CSVD measures (e.g., WMH volume/score), we pooled standardized mean differences (Hedges g) with 95% CIs. Ordinal CSVD scales (e.g., Fazekas) were analyzed as continuous variables, an approach commonly used in imaging meta-analyses that assumes approximately equal spacing between categories and maximizes use of available information. For dichotomous outcomes, we pooled log odds ratios (applying a 0.5 continuity correction for zero cells) and present results as odds ratios. Directions were harmonized so that positive values indicate greater CSVD burden in PD. Between-study heterogeneity was assessed with Cochran Q test ($p < 0.10$ considered significant) and quantified using I^2 (low 0%–40%, moderate 30%–60%, substantial 50%–90%, considerable >75%) and τ^2. If significant heterogeneity was present ($I^2 \geq 50$ or $p < 0.1$), the random-effects model was selected.

Subgroup Comparison

In cases where studies reported outcomes separately for clinical subgroups (e.g., PD with and without REM sleep behavior disorder), we combined these subgroups to facilitate pairwise comparisons with controls, in line with PRISMA and Cochrane recommendations for data synthesis across independent subgroups.[16,17]

Assessment of Small Study Effects

We evaluated small study effects for pooled analyses with k > 10 using funnel plots, Egger regression test ($p < 0.10$), and the trim-and-fill method to estimate adjusted effect sizes. We report both the original and bias-adjusted pooled effects with their 95% CIs.

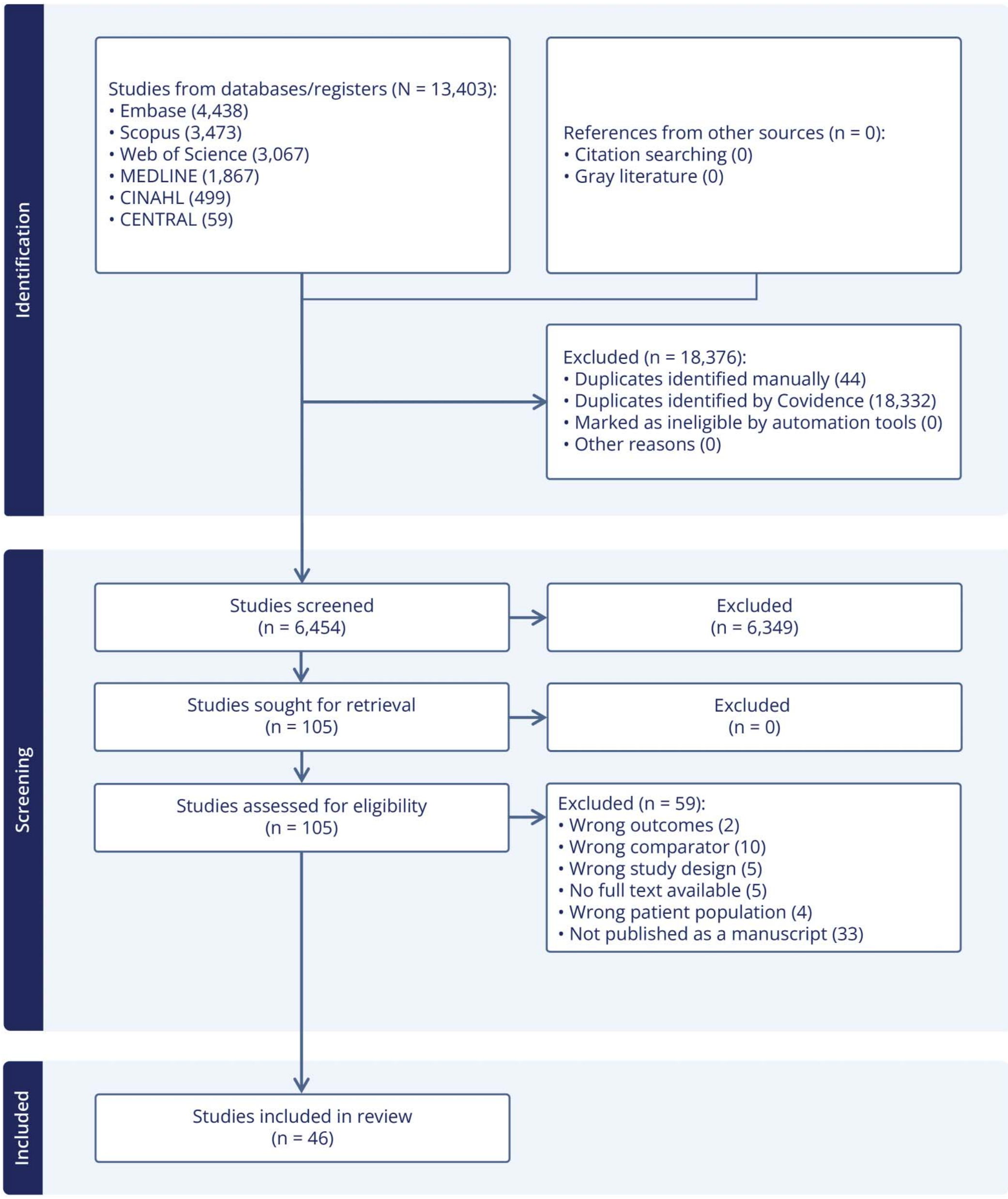
Sensitivity Analyses

We conducted sensitivity analyses iteratively excluding the following studies: those at high risk of bias defined as ROB <6 on the NOS, studies excluding participants based on cognitive impairment if impairment was defined as less than severe (MoCA >10) or excluding patients with baseline CSVD burden, those that either did not report MRI field strength or had a field strength of 1.5T or less, and studies that did not control for age and at least one vascular risk factor. An additional sensitivity analysis was performed to account for studies with potentially overlapping cohorts. When overlap was suspected (e.g., ≥2 shared authors) and cohort independence could not be determined from the reported recruitment frame and period, only one dataset per cohort was retained based on recency and cohort completeness.

Software

All analyses were performed in R (version 4.4.2) using the meta, metabias, and metafor packages. Forest plots, funnel plots, and statistical tests were generated accordingly.

Figure 1 PRISMA Flow Diagram



CENTRAL = Cochrane Central Register of Controlled Trials; CINAHL = Cumulative Index to Nursing and Allied Health Literature.

Reporting

Results are reported in accordance with PRISMA guidelines. Effect sizes are presented with 95% confidence intervals (CIs) and p values; heterogeneity statistics (I^2, Q, p values) are provided for each pooled analysis.

Standard Protocol Approvals, Registrations, and Patient Consents

Institutional review board approval and patient consent were not required for this study because it is a systematic review and meta-analysis of previously published literature and did not involve direct interaction with human subjects or access to identifiable patient data.

Data Availability

Data not published within this article will be made available by request from any qualified investigator.

Results

We identified 6,454 reports, screened 105 full texts, and included 46 studies in our initial search (Figure 1). Of these, 44 were cross-sectional and had data eligible for inclusion in at least one of the meta-analyses, encompassing a total of 3,817 people with PD and 2,593 HC (mean ages: 66.9 and 66.5, respectively). A total of 22 studies used volumetric measurements to compare people with PD and healthy controls, and one additional study was identified with an updated search. Characteristics of the 22 studies using volumetric measures can be found in eTable 1. There were 27 studies that used visual rating scales. Among them, 18 assessed global WMH, 11 assessed periventricular WMH (pvWMH), 11 assessed deep WMH (dWMH), 8 assessed CMB, 7 assessed lacunes, and 9 assessed ePVS. Burden of volumetric WMH, global WMH, pvWMH, dWMH, and ePVS in the midbrain were all significantly associated with PD. Risk of bias (RoB) was mixed across studies. Most studies showed low risk in representativeness, exposure ascertainment, outcome assessment, and statistical methods. However, sample size and nonrespondent reporting were consistently high risk, and control selection was frequently unclear or high risk. RoB figures are available in the supplement as eFigures 1–6. Overall evidence quality was rated as low because of the observational design of included studies. Heterogeneity, which was greatest for ePVS, did not warrant further downgrading given consistent results across sensitivity analyses summarized in eTable 2 and eFigures 7 and 8. Indirectness concerns were limited, as exclusions of patients with cognitive impairment or baseline CSVD would likely bias effects toward the null.

Kent et al.[18] was the only non–cross-sectional study included; its design precluded inclusion in the meta-analysis. They used natural language processing to identify WMH in radiology reports of >230,000 individuals ≥50 years old with nonstroke neuroimaging (2009–2019) and followed them starting 180 days postscan to determine whether they developed PD by billing codes. WMD was associated with an increased risk of developing PD (HR 2.10, 95% CI 1.92–2.29). Compta et al.[19] compared ARWMC regional scores and, therefore, not amenable to inclusion in the global WMH, pvWMH, or dWMH meta-analyses. They separated the PD cohort into those with and without dementia, finding those with dementia to have a significantly greater proportion of moderate-to-severe WMHs, defined as an ARWMC score ≥2, in the parieto-occipital region (p = 0.023).

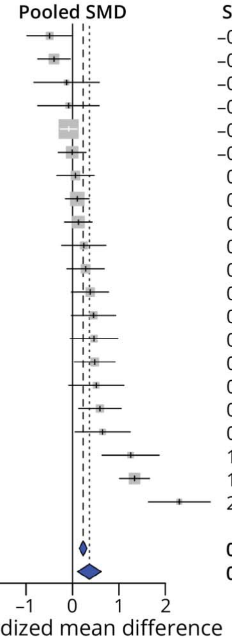
WMH Volumetric Measurements

Of 23 studies, 21 reported total WMH burden suitable for comparison. Comparing these studies with a random-effects model (I^2 = 85.6%, $p < 0.0001$) yielded a pooled standardized mean difference (SMD) of 0.36 (95% CI 0.11 to 0.62) (Figure 2). Egger regression provided evidence of small study effects (t = 1.95, df = 19, p value = 0.0655), and visual inspection of funnel plot shown in eFigure 9 suggested asymmetry. Using trim-and-fill, 6 studies were imputed (increasing k to 27), resulting in an adjusted pooled SMD of 0.08 (95% CI −0.24 to 0.39). Dadar et al[20]. 2022 was excluded from pooling because of lacking suitable total WMH volume measures and ultimately reported null findings in the PPMI cohort. Reheman et al[21]. 2025 compared WMH volumes among patients with PD and healthy controls, and although there was a trend toward higher WMH in the PD group, this was not statistically significant (p = 0.153).

Figure 2 Meta-Analysis of WMH Volumes in Parkinson Disease vs Healthy Controls

Study	Parkinson disease N	Parkinson disease Mean	Parkinson disease SD	Healthy controls N	Healthy controls Mean	Healthy controls SD	SMD	95% CI	Wt (random) (%)
Rane 2020[e20]	41	2.98	1.5670	28	3.92	2.2750	−0.49	(−0.98; −0.01)	4.7
Jones 2017[e21]	67	0.01	0.0100	61	0.01	0.0100	−0.40	(−0.75; −0.05)	5.1
Ball 2018[e10]	15	5.29	6.2000	16	6.49	11.3000	−0.13	(−0.83; 0.58)	3.9
Dunet 2019[e1]	15	8.20	7.1000	21	9.20	13.5000	−0.09	(−0.75; 0.58)	4.1
Dadar, Fereshtehnejad 2020[e3]	365	6.93	8.0300	174	7.66	10.3800	−0.08	(−0.26; 0.10)	5.5
Chahine 2019[e18]	141	10.67	5.2600	63	10.71	6.3600	−0.01	(−0.30; 0.29)	5.3
Dadar, Gee 2020[e2]	50	1.36	1.5700	45	1.26	1.4800	0.06	(−0.34; 0.47)	5.0
Dalakaer 2009[e4]	155	0.60	1.0000	98	0.50	0.9000	0.10	(−0.15; 0.36)	5.4
Dadar, Mahmoud 2022[e15]	76	9.97	9.4100	105	8.81	8.8100	0.13	(−0.17; 0.42)	5.3
Pozorski 2019[e11]	29	1.05	0.3170	42	0.98	0.2600	0.24	(−0.23; 0.72)	4.7
Grey 2022[e5]	42	2.93	5.3400	55	1.82	2.2500	0.28	(−0.12; 0.69)	5.0
Wu 2023[e7]	50	1.35	0.3800	47	1.21	0.3500	0.38	(−0.02; 0.78)	5.0
Gallagher 2017[e13]	29	0.37	0.6700	43	0.17	0.1300	0.45	(−0.02; 0.93)	4.7
Dunet 2017[e19]	32	0.91	0.8200	29	0.55	0.7000	0.46	(−0.05; 0.97)	4.6
Al-Bachari 2017[e12]	51	4.00	6.6667	34	1.41	2.2222	0.48	(−0.04; 0.92)	4.8
Dunet 2016[e17]	22	0.96	0.8700	23	0.55	0.7000	0.51	(−0.08; 1.11)	4.3
Al-Bachari 2020[e9]	49	1.80	0.9500	31	1.26	0.8300	0.59	(0.13; 1.05)	4.8
Liu 2015[e8]	23	5,827.00	10,755.0000	23	776.00	1,358.0000	0.65	(0.05; 1.24)	4.3
Fang 2021[e16]	19	1.75	1.4760	33	0.51	0.4950	1.25	(0.64; 1.87)	4.2
Scamarcia 2022[e6]	154	484.53	74.3400	60	388.89	62.8200	1.34	(1.01; 1.66)	5.2
Fang, Ann 2020[e14]	44	0.01	0.0020	20	0.00	0.0020	2.30	(1.63; 2.97)	4.1
Common effect model	**1,469**			**1,051**			**0.23**	**(0.15, 0.31)**	**.**
Random effects model							**0.36**	**(0.11, 0.62)**	**100.0**

Heterogeneity: I^2 = 85.6%, τ^2 = 0.2970, $p < 0.0001$



Forest plot of SMD comparing total WMH volume between individuals with Parkinson disease (PD) and healthy controls across studies that used volumetric MRI quantification. Positive SMD indicates greater WMH volume in PD. Higher WMH volume burden is associated with PD. PD = Parkinson disease; SMD = standardized mean difference; WMH = white matter hyperintensities.

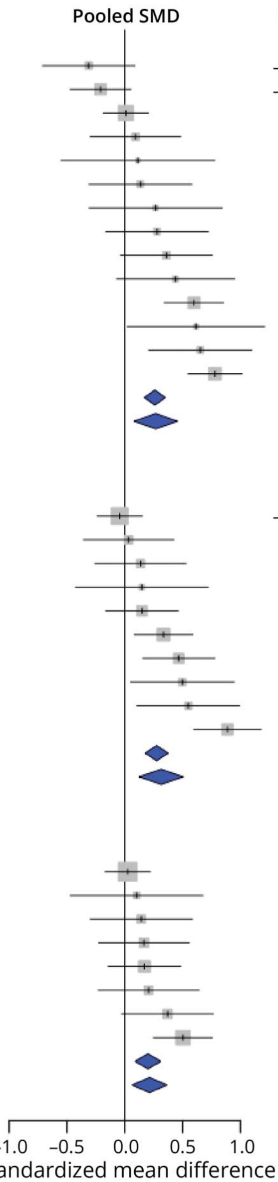
Visual Rating Scales (VRS)

Global WMH

Eighteen studies compared global WMH, and 14 were suitable for meta-analysis. Pooled SMD across the 14 studies from a random effects model ($I^2 = 78.1\%$, $p < 0.0001$) was 0.27 (95% CI 0.08–0.46) (Figure 3). Egger regression test did not show evidence of small study effects ($t = 0.15$, $df = 12$, $p = 0.88$); however, there was a suggestion of asymmetry on visual inspection of the funnel plot shown in eFigure 10. Using trim-and-fill, 6 studies were imputed (increasing k to 20), resulting in an adjusted pooled SMD of 0.01 (95% CI −0.22 to 0.23). We excluded Tu et al.[22] because of internal inconsistency (identical group medians/IQR despite $p < 0.001$). The other 3 studies not included only had prevalence data with cutoffs and, therefore, were not suitable for comparison: Tsai et al. found similar total Fazekas ≥2 prevalence in PWP and HC (28.1% vs 29.4%, $p = 1$), Al Bacchari et al. (2014) with a trend of higher prevalence of Wahlund score ≥2 in the PD group (42.9% vs 14.3%, $p = 0.2$), and Linortner with a trend of higher prevalence of either pvWMH score of 3 or dWMH score ≥2 in the PD group (19.7% vs 5.3%, $p = 0.731$).[23-25] Individual study methodological details are summarized in eTable 3.

Figure 3 Meta-Analysis of Global and Regional WMH Burden in Parkinson Disease vs Healthy Controls

Study	N	Mean	SD	N	Mean	SD	SMD	95% CI	Wt (common) (%)	Wt (random) (%)
Region = Global WMH										
Si 2020[e26]	82	1.85	1.3500	35	2.30	1.6000	−0.31	(−0.71; 0.08)	5.2	7.3
Gu 2023[e28]	124	1.33	1.7400	106	1.67	1.4800	−0.21	(−0.47; 0.05)	12.1	8.9
Kim 2018[e34]	205	1.95	2.1500	205	1.93	2.0400	0.01	(−0.18; 0.20)	21.9	9.6
Meng 2023[e32]	51	1.31	1.4600	50	1.16	1.7400	0.09	(−0.30; 0.48)	5.4	7.3
Dunet 2019[e1]	15	1.50	0.8000	21	1.40	0.9000	0.11	(−0.55; 0.78)	1.9	4.6
Acharya 2007[e23]	36	3.70	4.2000	43	3.20	3.1000	0.14	(−0.31; 0.58)	4.2	6.7
Alves 2022[e30]	24	2.17	1.2100	23	1.85	1.1400	0.27	(−0.31; 0.84)	2.5	5.4
Petrovic 2012[e33]	59	8.14	7.3600	30	6.20	5.8000	0.28	(−0.16; 0.72)	4.2	6.8
Joki 2018[e35]	50	2.78	0.8000	50	2.48	0.8500	0.36	(−0.03; 0.76)	5.2	7.3
Dunet 2017[e19]	32	1.80	0.9000	29	1.40	0.9000	0.44	(−0.07; 0.95)	3.2	6.0
Tokuchi 2016[e36]	153	2.70	1.4900	103	1.90	1.0600	0.60	(0.34; 0.85)	12.6	8.9
Liu 2015[e8]	23	5.80	5.3000	23	3.10	3.0000	0.62	(0.02; 1.21)	2.3	5.2
Al-Bachari 2017[e12]	51	8.37	6.5500	34	4.52	4.5500	0.65	(0.21; 1.10)	4.1	6.7
Song 2017[e31]	241	1.23	0.8300	112	0.60	0.7500	0.78	(0.55; 1.01)	15.3	9.2
Common effect model	.			.			**0.26**	**(0.17; 0.35)**	**100.0**	.
Random effects model							**0.27**	**(0.08; 0.46)**	.	**100.0**
Heterogeneity: $I^2 = 78.1\%$, $\tau^2 = 0.0885$, $p < 0.0001$										
Region = pvWMH										
Kim 2018[e34]	205	0.99	0.7400	205	1.02	0.6600	−0.04	(−0.24; 0.15)	26.4	13.1
Meng 2023[e32]	51	0.67	0.7700	50	0.64	0.9800	0.03	(−0.36; 0.42)	6.5	9.3
Joki 2018[e35]	50	1.40	0.5300	50	1.32	0.6200	0.14	(−0.25; 0.53)	6.4	9.3
Alves 2022[e30]	24	1.16	0.8000	23	1.04	0.8000	0.15	(−0.43; 0.72)	3.0	6.5
Ma 2021[e39]	107	.	.	62	.	.	0.15	(−0.16; 0.46)	10.1	10.8
Tokuchi 2016[e36]	153	1.40	1.0000	103	1.10	0.7000	0.34	(0.08; 0.59)	15.6	12.0
Piccini 1995[e38]	102	0.67	0.9200	68	0.24	0.9500	0.47	(0.16; 0.78)	10.2	10.9
Petrovic 2012[e33]	59	1.79	2.1698	30	0.80	1.5000	0.50	(0.05; 0.94)	5.0	8.4
Al-Bachari 2017[e12]	151	1.53	1.0200	34	0.98	0.9500	0.55	(0.11; 0.99)	5.1	8.4
Ma 2023[e37]	100	1.33	0.7400	100	0.67	0.7400	0.89	(0.60; 1.18)	11.7	11.3
Common effect model	.			.			**0.28**	**(0.18; 0.38)**	**100.0**	.
Random effects model							**0.32**	**(0.12; 0.51)**	.	**100.0**
Heterogeneity: $I^2 = 73.8\%$, $\tau^2 = 0.0639$, $p < 0.0001$										
Region = dWMH										
Kim 2018[e34]	205	0.96	2.0200	205	0.91	1.9300	0.03	(−0.17; 0.22)	33.8	23.0
Alves 2022[e30]	24	1.01	2.0800	23	0.81	1.6900	0.10	(−0.47; 0.68)	3.9	5.8
Petrovic 2012[e33]	59	6.35	7.0300	30	5.40	5.6000	0.14	(−0.30; 0.58)	6.5	8.9
Meng 2023[e32]	51	0.65	0.7400	50	0.52	0.8100	0.17	(−0.22; 0.56)	8.3	10.6
Ma 2021[e39]	107	.	.	62	.	.	0.17	(−0.14; 0.48)	12.9	14.2
Al-Bachari 2017[e12]	51	1.32	2.7200	34	0.82	1.7100	0.21	(−0.23; 0.64)	6.7	9.0
Joki 2018[e35]	50	1.38	0.6000	50	1.16	0.5800	0.37	(−0.03; 0.77)	8.1	10.4
Tokuchi 2016[e36]	153	1.30	1.1000	103	0.80	0.8000	0.50	(0.25; 0.76)	19.7	18.1
Common effect model	.			.			**0.20**	**(0.09; 0.31)**	**100.0**	.
Random effects model							**0.21**	**(0.06; 0.36)**	.	**100.0**
Heterogeneity: $I^2 = 26.9\%$, $\tau^2 = 0.0159$, $p = 0.2143$										



Forest plot of SMD comparing WMH burden between Parkinson disease (PD) and healthy controls across 3 regions: Global WMH, pvWMH, and dWMH. Positive SMD indicates greater WMH burden in PD. Higher WMH burden as measured using VRS for global, periventricular and deep regions, was associated with PD. dWMH = deep WMH; PD = Parkinson disease; pvWMH = periventricular WMH; SMD = standardized mean difference; WMH = white matter hyperintensities.

Periventricular WMH

Eleven studies evaluated pvWMH with a VRS, details in eTable 4. Al Bachari et al. (2014) reported prevalence alone with a trend toward higher pvWMH prevalence in PD population (p = 0.2), and the remaining 10 were included in a meta-analysis.[24] Pooled SMD from a random effects model (I^2 = 73.8%, p < 0.0001) was 0.32 (95% CI: 0.12–0.51) (Figure 3). Egger regression test did not show evidence of small study effects (t = 0.87, df = 8, p = 0.41), and the funnel plot was symmetric on visual inspection, shown in eFigure 11. After trim-and-fill, 2 studies were imputed (increasing k to 12), resulting in an adjusted pooled SMD of 0.20 (95% CI −0.02 to 0.42).

Deep WMH

Eleven studies evaluated dWMH VRS separately, and 8 were included in a meta-analysis; methodological details available in eTable 4. The pooled SMD from a common effects model (I^2 = 26.9%, p = 0.214) was 0.20 (95% CI 0.09–0.31) (Figure 3). Ma 2023 et al. was excluded from analysis because of internal inconsistency (identical group medians/IQR despite p < 0.001).[26] Al Bachari 2014 assessed prevalence alone with a trend toward higher dWMH prevalence in PD population (35.7% vs 14.3%, p = 0.4).[24] Piccini 1995 similarly reported prevalence only and found PWP had a dWMH prevalence of 35.2% compared with 41.1%, which was not statistically significant.[27]

Cerebral Microbleeds

Eight studies assessed CMB burden with study details presented in eTable 5. Cumulative OR for the 6 studies reporting CMB prevalence using a random effects model (I^2 = 69.0%, p = 0.0065) was 1.18 (95% CI 0.57–2.42) (Figure 4). We excluded Tu et al.[22] because of internal inconsistency (identical group medians/IQR despite p < 0.001). Ma 2021 reported nonsignificant results but did not share the individual values or measures of association.[9]

Lacunar Infarcts

Seven studies assessed lacunar infarct burden; however, 3 of which reported number of lacunes rather than group prevalence and, therefore, not amenable for comparison. Study details summarized in eTable 6. The remaining 4 studies reporting prevalence values were summarized using a random effects model (I^2 = 67.9%, p = 0.0252) resulting in a cumulative OR of 1.48 (95% CI 0.58–3.78) (Figure 5). Tu et al. reported greater LI number burden in PWP (p < 0.001); Gu et al. and Ma et al. 2021 did not find significant differences.[9,22,28]

Enlarged Perivascular Spaces

EPVS were measured in 9 studies, 6 of which quantified ePVS in the centrum semiovale (CSO), 4 in the midbrain (MB), and 7 in the basal ganglia (BG). A summary of study details can be found in eTable 7. Fang, Gu 2020 et al. was the only study among the 9 not included in the meta-analyses as they only reported single prevalence cutoffs; they found higher BG ePVS prevalence in PWP compared with HC (p = 0.001) and no significant differences for either the MB ePVS or the CSO ePVS.[29] Each regional subgroup was pooled with a random effects model because of significant heterogeneity. The pooled SMD for the remaining 5 studies comparing ePVS in CSO was 1.02 (95% CI: −0.14-2.18) (Figure 6). The pooled SMD for the remaining 3 measuring ePVS in the MB resulted in an SMD of 1.8 (95% CI: 0.15–3.44) (Figure 6). Finally, the remaining 6 studies comparing ePVS burden in the BG were pooled with a cumulative SMD of 1.54 (95% CI: −0.30-3.39) (Figure 6).

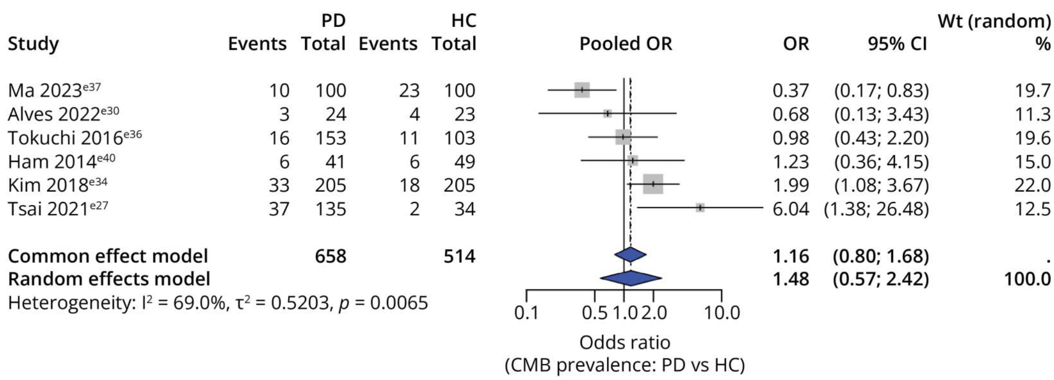
Cumulative CSVD Burden

Four studies measured a total CSVD burden score to compare across patients with PD and HC. Methodological details summarized in eTable 8. Kim et al. and Ma 2021 et al. calculated a total CSVD score and found no significant difference between PWP and HC.[9,30,31] Patel et al. (CT/MRI) and Song et al. (MRI) reported higher burden in PD (OR 2.2, p = 0.02 and 75.5% vs 43.8%, p < 0.001, respectively).[32,33]

Sensitivity Analyses

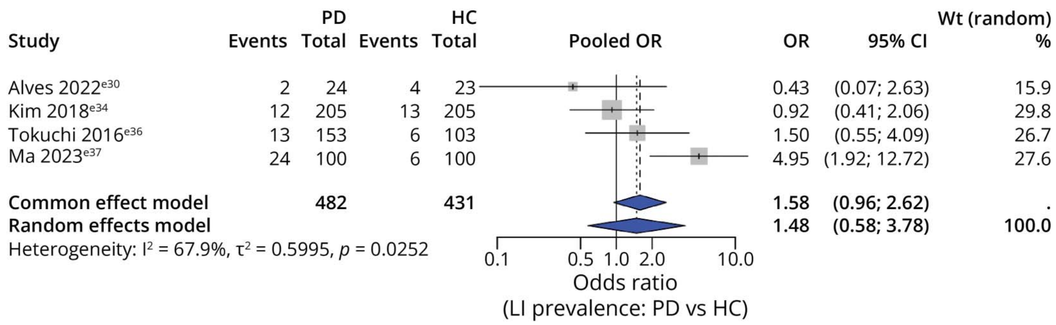
A summary of the results of the sensitivity analyses can be found in eTable 2 and eFigures 7 and 8. Exclusion of studies based on higher risk of bias had the tendency to increase the

Figure 4 Meta-Analysis of CMB Prevalence in Parkinson Disease vs Healthy Controls



Forest plot of OR comparing CMB prevalence in PD vs HC. OR > 1 indicates greater CMB prevalence in PD. No significant differences appreciated between PD and HC. CMB = cerebral microbleed; PD = Parkinson disease; SMD = standardized mean difference; WMH = white matter hyperintensities.

Figure 5 Meta-Analysis of Lacunar Infarct Prevalence in Parkinson Disease vs Healthy Controls



Forest plot of OR comparing lacunar infarct prevalence in PD vs HC. OR > 1 indicates greater lacunar infarct prevalence in PD. No significant differences appreciated between PD and HC. LI = lacunar infarct; PD = Parkinson disease; SMD = standardized mean difference; WMH = white matter hyperintensities.

size of the confidence interval without meaningfully changing results. Similarly, exclusion of studies excluding either mild to moderate cognitive impairment (MoCA >10) or baseline CSVD burden did not significantly affect results. Although most CSVD features were not affected by sensitivity analyses considering only 3T scanners, BG ePVS burden reached statistically significant differences with an SMD of 0.62 (95% CI 0.09–1.15). Twenty-three studies had potential participant overlap, comprising 6 groups with ≥2 shared authors. When overlap was suspected, one study per group was retained based on recency or methodological considerations, while explicitly independent cohorts were retained. A breakdown of the potentially overlapping cohorts in each CSVD feature and studies retained is provided in eTable 9. This duplicate-cohort sensitivity analysis did not materially alter the corresponding results. Excluding studies that failed to control for age and at least one vascular risk factor also left most findings unchanged. The only exception was cerebral microbleeds, where restricting to adequately controlled studies revealed a significantly higher CMB burden in PD compared with controls (OR 1.89, 95% CI 1.16–3.07).

Discussion

This review supports the hypothesis that CSVD is more common in PD than in healthy controls. Across multiple studies, PD was significantly associated with greater WMH burden. This was observed with both volumetric analyses and visual rating scales, and in global, periventricular, and deep regions. Associations based on volumetric and global WMH measures lost significance after trim-and-fill correction, which may reflect publication bias, small study effects, or the substantial heterogeneity observed. The midbrain ePVS were also found to be significantly associated with PD and in excluding studies using less the 3T MRI field strength, basal ganglia ePVS burden in PD also reached statistical significance. CMB prevalence in PD trended higher than HC and reached significance with the exclusion of studies that failed to control for age and at least one vascular risk factor. Although lacunar infarct prevalence and centrum semiovale ePVS burden did not reach statistical significance, both showed a trend toward higher values in the PD cohort. The lack of significance may reflect limited power and substantial heterogeneity across studies. The robustness of results across the sensitivity analyses demonstrates the strength of these findings, which suggest that NVU dysfunction may disproportionately predispose individuals with PD to these MRI features.

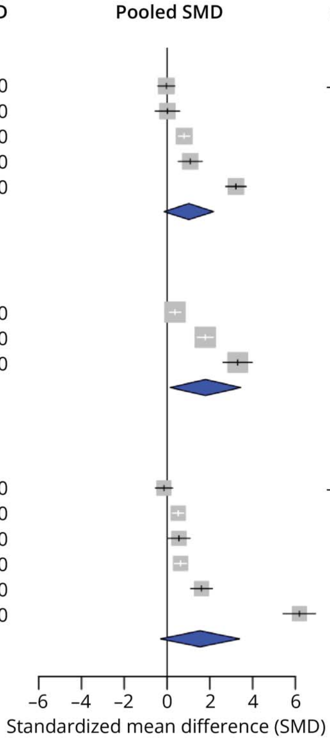
Whether NVU dysfunction precedes PD onset or whether CSVD predisposes individuals to PD remains unresolved. Kent et al.[18] found a surprisingly high hazard ratio of 2.1 for developing PD among patients incidentally found to have white matter disease, which remained significant even when restricting analysis to those who developed PD > 1 year after imaging. Given the prodromal phase of PD can span years before motor diagnosis, reverse causality cannot be excluded. In addition, the outcome of PD was determined by International Classification of Diseases, 10th Revision code, raising the possibility of mixed PD cohorts including vascular Parkinson disease and other causes of parkinsonism. Similarly, van der Holst et al.[34] found that high WMH volume and high lacune number predicted incident parkinsonism within 5 years. Although these studies reinforce an association between PD and CSVD, they do not clarify directionality or causality. This is also true for the present study, which by its observational and primarily cross-sectional nature is not able to clarify the causal direction of the potential association between PD and CSVD.

Across included studies, the high risk-of-bias ratings for sample size and nonrespondent domains likely reflect practical limitations rather than intentional sampling strategies, as most studies were secondary analyses of existing PD cohorts rather than cohorts designed specifically for CSVD research. Notably, restricting analyses to studies that appropriately controlled for age and vascular risk factors produced largely unchanged effect estimates across all CSVD markers, indicating that these methodological weaknesses did not meaningfully alter the overall conclusions.

The pathophysiologic mechanisms leading to CSVD are still under investigation. Although WMH are traditionally

Figure 6 Meta-Analysis of ePVS Burden in Parkinson Disease vs Healthy Controls

Study	Parkinson disease N	Mean	SD	Healthy controls N	Mean	SD	SMD	95% CI	Wt (random) (%)
Region = CSO									
Si 2020[e26]	82	16.68	9.5900	35	17.00	7.9000	−0.03	(−0.43; 0.36)	20.2
Alves 2022[e30]	24	1.36	1.1800	23	1.34	1.2300	0.02	(−0.56; 0.59)	19.7
Gu 2023[e28]	124	17.62	7.8600	106	11.98	5.8100	0.80	(0.53; 1.07)	20.4
Tu 2022[e25]	26	17.58	10.1900	32	8.42	6.4800	1.08	(0.53; 1.64)	19.7
Li 2020[e41]	88	3.41	0.1280	71	3.01	0.1180	3.22	(2.74; 3.70)	20.0
Random effects model	**344**			**267**			**1.02**	**(−0.14; 2.18)**	**100.0**
Heterogeneity: $I^2 = 96.7\%$, $\tau^2 = 1.6902$, $p < 0.0001$									
Region = MB									
Gu 2023[e28]	124	2.90	1.8900	106	2.21	1.7900	0.37	(0.11; 0.63)	34.0
Li 2020[e41]	**88**	2.33	0.2390	71	1.86	0.2910	1.78	(1.42; 2.16)	33.7
Shen 2021[e43]	**40**	8.20	0.6000	41	6.20	0.6000	3.30	(2.62; 3.98)	32.4
Random effects model	**252**			**218**			**1.80**	**(0.15; 3.44)**	**100.0**
Heterogeneity: $I^2 = 97.6\%$, $\tau^2 = 2.0636$, $p < 0.0001$									
Region = BG									
Si 2020[e26]	82	11.52	6.7400	35	12.40	3.7000	−0.15	(−0.54; 0.25)	**16.7**
Gu 2023[e28]	124	9.96	5.8700	106	7.15	4.7800	0.52	(−0.26; 0.78)	**16.8**
Tu 2022[e25]	26	8.58	5.0000	32	6.33	2.9600	0.55	(0.03; 1.08)	**16.6**
Ma 2023[e37]	100	1.25	0.5600	100	1.00	0.0100	0.63	(0.34; 0.91)	**16.8**
Shen 2021[e43]	40	12.40	0.8000	41	11.10	0.8000	1.61	(1.11; 2.11)	**16.7**
Li 2020[e41]	88	1.81	0.0770	71	1.35	0.0700	6.19	(5.43; 6.95)	**16.4**
Random effects model	**460**			**385**			**1.54**	**(−0.30; 3.39)**	**100.0**
Heterogeneity: $I^2 = 97.9\%$, $\tau^2 = 5.2713$, $p < 0.0001$									



Forest plot of SMD comparing EPVS burden in Parkinson disease vs healthy controls across 3 anatomical regions: CSO, MB, and BG. Positive SMD indicates greater EPVS burden in PD. Higher ePVS burden in the MB was associated with PD. BG = basal ganglia; CSO = centrum semiovale; ePVS = enlarged perivascular spaces; HC = healthy controls; MB = midbrain; PD = Parkinson disease; SMD = standardized mean difference.

associated with ischemic injury, they also reflect decreased nerve fiber and myelin density and reactive gliosis.[35] MB ePVS may reflect impaired fluid clearance or glymphatic dysfunction, further linking NVU compromise with downstream structural changes.[3] It is well accepted that the WMH seen in CSVD are frequently sequelae of ischemic insults, which may be enhanced by neurovascular coupling dysfunction in PD.[36-38] Alternatively, NVU breakdown may result in BBB permeability changes, leading to accumulation of neurotoxic metabolites and abnormal fluid shifts.[39-43] Decreased glymphatic system activity and impaired meningeal lymphatic drainage in PD may predispose to increased transependymal flow and accumulation of toxic substances observed on MRI as pvWMH and ePVS.[3,28,44-47]

One potential confounder in these studies is the frequent exclusion of patients with PD with extensive cerebrovascular disease under the assumption that vascular pathology may be causing secondary parkinsonism. Although reasonable for defining idiopathic PD cohorts, excluding patients with PD with the most extensive burden of CSVD may lead to a reduction of the effect size of this association. Another patient characteristic that was frequently excluded from general PD cohorts was mild to moderate cognitive impairment. This was the case, for example, in the large COMPASS-ND analysis by Dadar et al. who reported no significant WMH differences between cognitively intact PD and controls. Given that CSVD is associated with worse cognition, this exclusion may reduce the effect size. This same large cohort null study adjusted only for age and sex in their regression models (Methods, p. 4) and did not account for major vascular risk factors. Because CSVD burden increases with vascular risk exposure, incomplete adjustment in large observational cohorts can dilute group differences. This helps explain why some large datasets report null findings and underscores the importance of vascular matching in CSVD research in PD. To mitigate the mediator bias introduced from cognitive impairment, a prospective longitudinal cohort study enrolling patients with early or prodromal PD alongside matched healthy controls and tracking the incidence and progression of CSVD markers would be ideal. In addition, more refined biomarkers of disease are needed to distinguish other causes of parkinsonism from idiopathic PD with comorbid CSVD.

PD-related susceptibility to CSVD may be explained by a number of the aforementioned disruptions in the NVU and ultimately potentiate clinical deterioration. CSVD, therefore, represents a potentially modifiable pathway of disease progression in PD. Aerobic exercise, already endorsed in PD care, has the potential to reduce progression of CSVD and has already been shown to improve outcomes in PD.[48,49] Recent work by Jung et al.[50] observed that dihydropyridine use was associated with a lower risk of dementia conversion in patients with PD, a finding potentially mediated by CSVD attenuation, although causal inference requires randomized trials. Interventions aimed at improving sleep regulation and glymphatic function may offer additional avenues worth exploring. More research is required to determine the degree and characteristics of CSVD among patients with Parkinson disease with improved biomarkers and more thoughtful inclusion/exclusion criteria. More inclusive cohort criteria and harmonized outcome measures will be essential to delineate to what extent these characteristics represent biomarkers of NVU dysfunction, and what interventions if any may be used to target these important pathways and slow disease progression.

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Author Contributions

D.F. Del Papa: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. J. Herman: major role in the acquisition of data. A.T. Kunju: major role in the acquisition of data. T. Najafi: major role in the acquisition of data. R. O'Boyle: major role in the acquisition of data. A. Salama: major role in the acquisition of data. L. Hneiny: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. T. Samarakoon: drafting/revision of the manuscript for content, including medical writing for content. T. Peabody: major role in the acquisition of data. S. Marmol: major role in the acquisition of data. M. Feldman: drafting/revision of the manuscript for content, including medical writing for content. I.U. Haq: Drafting/revision of the manuscript for content, including medical writing for content; study concept or design.

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